

Contradiction Fields: CRT Rigidity, Prime Matrix Geometry, and Zeta-Zero Escape

Project manuscript

May 3, 2026

Abstract

This monograph draft unifies two contradiction-field research programs. The first studies prime existence in CRT prime matrices through row, column, diagonal, anchor, and rough-composite rigidity. The second studies an off-critical zero of the Riemann zeta function through a global contradiction-field ledger for prime anomalies. The common method is to convert a hypothetical counterexample into a structured covering field, decompose it into local rigidity layers, and exclude all free escape channels by capacity, periodicity, and no-cycle bookkeeping.

This is a synthesis and dependency manuscript. It does not assert that the Riemann Hypothesis or the prime-matrix row/column theorem has already been accepted as a final unconditional theorem. Each claim is marked by status in the accompanying status table: proved in manuscript, reduced to a structured input, external, computationally certified, or requiring independent referee verification.

Contents

1	Guiding Principle: The Contradiction Field	7
2	Prime-Matrix Contradiction Fields	9
2.1	Two-Point Rough-Pair Propositions	9
2.2	Two-Point Secondary-Sieve Contradiction Field	10
3	Zeta-Zero Contradiction Fields	17
4	Unified Dependency Map	19
5	Directory and Theory System	21
5.1	Proposed Book Structure	21
5.2	Latest Two-Point Chain	21
5.3	Prime-Matrix Low-Hole Closure Package	22
5.4	Claim Status Legend	22
5.5	External Theorem Index	23
5.6	Critical Chain Audit	23
5.7	Optimization Priorities	23
5.8	Pre-External-Referee Hard Obligations	24
6	Referee Closure Audit	29
7	Current Status and Honest Boundary	31

Chapter 1

Guiding Principle: The Contradiction Field

A contradiction field is a structured ledger attached to a hypothetical counterexample. Instead of estimating a single exceptional set, it records every possible explanation for the counterexample as a named channel. A proof closes when every channel is either absorbed, forced to descend in a finite potential, transferred with source deletion, or contradicts a global capacity lower/upper bound.

The common rigidities used throughout this manuscript are:

1. CRT skeletons and nonzero-residue balance;
2. local short-window non-reuse of large prime factors;
3. small-prime shell migration and diagonal locking;
4. rough-composite and double-anchor decompositions;
5. capacity ledgers after baseline subtraction;
6. no-cycle potentials for internal escape mechanisms.

Chapter 2

Prime-Matrix Contradiction Fields

Let P be an odd prime and write the $P \times P$ prime matrix entries as

$$n(r, c) = (r - 1)P + c, \quad 1 \leq r, c \leq P.$$

The row and column programs attempt to show that every nontrivial row or column contains a prime. The current formal appendix reduces a row or non- P column counterexample to a structured bad configuration after removing small-factor locks and tail anchors.

Reduction-closed Statement 2.1 (A/B reduction, recorded form). *A nontrivial full-composite row or column, after small-factor and tail-anchor peeling, induces a Structured-EHPD bad configuration satisfying CRT balance, large-factor non-reuse, and anchor-capacity constraints.*

Reference. See `docs/row-column-reduction-formal-appendix.md` and the theorem list

`docs/monograph/prime-matrix-ab-entrance-theorem-list.md`.

This statement is a reduction interface; it does not by itself prove the Structured-EHPD exclusion theorem. $\square$

2.1 Two-Point Rough-Pair Propositions

A natural next strengthening asks for two simultaneous nonzero-residue constraints in the back half of the prime matrix. Fix an even integer $w > 0$ and ask whether there is an entry x in the back half of the $P \times P$ matrix such that both x and $x - w$ are nonzero modulo every prime $p \leq P$, or equivalently both x and $w - x$ are nonzero modulo every prime $p \leq P$.

Proposition 2.2 (Equivalence and strength of the two-point rough-pair problem). *For fixed even $w > 0$, the two conditions*

$$p \nmid x(x - w) \quad (p \leq P)$$

and

$$p \nmid x(w - x) \quad (p \leq P)$$

are identical. In the region $1 < x \leq P^2$, any integer nonzero modulo every prime $p \leq P$ is prime. Consequently, a proof of this two-point assertion for every sufficiently large P would imply infinitely many prime pairs of gap w ; in particular $w = 2$ would imply the twin-prime conjecture.

Proof. For each $p \leq P$, the condition $p \nmid x - w$ is $x \not\equiv w \pmod{p}$, and the condition $p \nmid w - x$ is the same congruence. Both formulations therefore exclude exactly the two residue classes 0 and w modulo each p , with possible collision when $p \mid w$.

If $1 < x \leq P^2$ is nonzero modulo every prime $p \leq P$ and is composite, then it has a prime factor $q \leq \sqrt{x} \leq P$, contradicting the nonzero-residue condition. Thus such an x is prime. If the two-point assertion holds for all sufficiently large $P > w$, then x and $x - w$ are both prime in infinitely many growing back-half matrix intervals, giving infinitely many prime pairs of gap w . $\square$

Remark 2.3 (Status). This two-point problem is therefore not a routine corollary of the one-point row/column contradiction field. It is a prime-pair-level strengthening, naturally expressible as a two-point rough Jacobsthal problem. In this monograph it is recorded as a future contradiction-field direction, not as an unconditional theorem.

2.2 Two-Point Secondary-Sieve Contradiction Field

The two-point rough-pair problem admits a more detailed contradiction-field research program. The current working manuscript is recorded in

`docs/monograph/two-point-secondary-sieve-research.md`.

It studies the secondary sieve condition

$$x \not\equiv 0, w \pmod{p} \quad (p \leq P)$$

inside back-half matrix windows, and tries to exclude a two-point rough-pair counterexample by a hierarchy of local rigidity fields.

Definition 2.4 (Two-point secondary-sieve field). Fix an even integer $w > 0$. A two-point secondary-sieve field is the structured ledger obtained from a hypothetical interval in which every candidate x fails at least one of the two congruence conditions $p \nmid x$ or $p \nmid x - w$ for some prime $p \leq P$. Its named channels are small-prime double locks, top-scale large-factor coverage, short-block zero-mass loss, multiplicative near-crossings, Fourier CRT defects, and smoothing boundary terms.

Lemma 2.5 (Fixed-order local crossing complexity). *In the secondary-sieve manuscript, every fixed-order local crossing test is expanded only through truncated hard-skeleton weights*

$$d \leq R, \quad R \leq (\log P)^{C_R},$$

with coefficient mass $\sum_{d \leq R} |\lambda_d| \leq (\log P)^{C_\lambda}$ and short-block shifts $|h| < L \asymp (\log \log P)^2$. For every fixed order k , the effective historical complexity satisfies

$$Q_{\text{eff},k} \ll_k \left(\sum_{d \leq R} |\lambda_d| \right)^k R^k L^{O(k)} \leq (\log P)^{C_k}.$$

In particular the required $k = 2, 3$ local crossing tests have polylogarithmic effective complexity.

Proof. After expanding the k truncated weights, only moduli $d_1, \dots, d_k \leq R$ occur. For fixed new-line primes p_i , phases, and block shifts, all conditions are one-variable linear congruences. By the Chinese remainder theorem the local system is either inconsistent or occupies one residue class modulo a divisor of

$$\text{lcm}(d_1, \dots, d_k) \text{lcm}(p_1, \dots, p_k).$$

The primes p_i are the explicit variables being summed in the crossing estimate; they contribute the usual Y^k scale, not hidden historical complexity. The historical part is bounded by $\text{lcm}(d_1, \dots, d_k) \leq R^k$, the block shifts contribute $L^{O(k)}$ patterns, and the coefficient expansion contributes $(\sum |\lambda_d|)^k$. This proves the displayed bound. The true residual set U_Y is not expanded into a full primordial period. $\square$

Lemma 2.6 (Zero-mass and smoothing discharge). *Assume the SC2/diagonal package supplies the moment bounds*

$$M_1 := \frac{\sum_B X_B R_B}{\sum_B X_B} \leq 0.45, \quad M_2 := \frac{\sum_B X_B \binom{R_B}{2}}{\sum_B X_B} \geq 0.05,$$

and

$$M_3 := \frac{\sum_B X_B \binom{R_B}{3}_+}{\sum_B X_B} \leq 0.03.$$

Then the weighted zero-hit mass satisfies

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \geq 0.57.$$

Moreover, if the directional-balance endpoint estimate gives

$$N_{\text{endpoint}}/N_{\text{total}} \leq C_{\text{end}}\Delta + o(1),$$

then choosing $\Delta \leq 0.01/C_{\text{end}}$ gives a smooth-to-sharp loss < 0.03 for all sufficiently large P .

Proof. For every integer $R \geq 0$,

$$1_{R=0} \geq 1 - R + \binom{R}{2} - \binom{R}{3}_+.$$

Multiplying by $X_B \geq 0$ and summing over blocks gives

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \geq 1 - M_1 + M_2 - M_3 \geq 1 - 0.45 + 0.05 - 0.03 = 0.57.$$

For smoothing, choose even smooth cutoffs $W_- \leq 1_{|t|<1} \leq W_+$ whose difference is supported in $1 - \Delta \leq |t| \leq 1 + \Delta$. The sharp and smoothed counts differ only on this endpoint layer. The displayed directional-balance bound therefore gives loss at most $C_{\text{end}}\Delta + o(1)$, which is < 0.03 after the stated choice of Δ and then taking P large. $\square$

Lemma 2.7 (Adaptive true-hit layering dichotomy). *Let U_Y be the current true residual set and define, for each new prime $p > Y$,*

$$a_p = \frac{1}{|U_Y|} \#\{x \in U_Y : x \equiv 0 \text{ or } w \pmod{p}\}.$$

Fix $\eta = 0.05$. Either some p has $a_p > \eta$, in which case the configuration is a Single-Prime CRTDefect, or the remaining primes can be greedily partitioned into consecutive layers $\mathcal{P}_j$ satisfying

$$\nu_j^{\text{real}} := \sum_{p \in \mathcal{P}_j} a_p \leq 0.4,$$

and, except for the final layer, $\nu_j^{\text{real}} > 0.4 - \eta \geq 0.35$.

Proof. If $a_p > \eta$ for some p , then a fixed positive proportion of U_Y lies in the two residue classes 0, w modulo p . This is exactly the Single-Prime CRTDefect exit, to be excluded by the CRTDefect/Directional Balance part of the SC2 package. Otherwise all $a_p \leq \eta$. Starting from the smallest remaining prime, add primes to the current layer until adding the next one would make the partial sum exceed 0.4. The completed layer has sum at most 0.4, and because the next summand is at most η , every non-final completed layer has sum greater than $0.4 - \eta$. This gives the claimed adaptive true-hit decomposition. $\square$

Proposition 2.8 (Recorded conditional closure chain for the two-point secondary sieve). *Using the three lemmas above, assume the following remaining input holds with the constants recorded in the secondary-sieve research note:*

1. *the I3-Core SC2/CRTDefect package controls the true-residual two-line near-crossing error, the odd-prime 45° main-synchronization peak, Single-Prime CRTDefect exits, the normalized $q = 2$ parity skeleton, the moment bounds M_1, M_2, M_3 , and the endpoint Directional Balance used in the zero-mass and smoothing discharge lemma.*

Then the two-point secondary-sieve contradiction field has no remaining top-scale escape channel under the parameter choice

$$\alpha_0 = 0.75, \quad c_0 = 1/2, \quad \nu_j^{\text{real}} \leq 0.4$$

for adaptive true-hit layers, subject to the same fixed- w convention used in the research manuscript.

Proof sketch and status. The argument is a conditional synthesis, not a final unconditional theorem. A top-scale covering excess is first decomposed into adaptive true-hit layers; any single-prime concentration is routed to the Single-Prime CRTDefect exit. The remaining excess is converted by a TCA linear bridge into a positive short-block load. The block load is forced into the SMC/SC2 near-crossing ledger. The 45° main synchronization is split into large-factor reuse, shared-complement factors, and good synchronized pairs; reuse and shared factors are excluded by short-window rigidity, while good pairs are sent to an auxiliary-prime Fourier CRT-defect. Directional balance excludes the Fourier defect unless the top-scale energy is already below the capacity threshold. Finally, the weighted Bonferroni zero-mass estimate supplies enough zero-hit block mass to make the capacity comparison incompatible.

The current audit is archived under `docs/archive/monograph-reviews/` and classifies this as a research proposition conditional on the single I3-Core package above. The fixed-order complexity input, zero-mass/smoothing discharge, and adaptive layering dichotomy have been inlined as the preceding lemmas. The fact that an unconditional proof would imply fixed-gap prime-pair consequences, including the twin-prime case when $w = 2$, is not a logical obstruction. It only marks the strength of the claim. The reason this chapter remains conditional is that I3-Core has not yet been inlined as an independently verified unconditional proof. $\square$

Proposition 2.9 (Window-compressed conditional variant). *Keep the same fixed even integer $w > 0$ and the same remaining I3-Core input as in the preceding proposition. Let I be any consecutive back-half window of $H(P)$ rows in the $P \times P$ matrix, so that $|I| = H(P)P + O(P)$. By the fixed-order complexity lemma, write*

$$Q_{\text{eff}} \leq C_Q (\log P)^C.$$

For a prescribed middle-scale error budget $\varepsilon_{\text{SC2}} > 0$, the conditional row threshold supplied by the current ledger is

$$H_{\min}^{\text{cond}}(P; \varepsilon_{\text{SC2}}) = \left\lceil \frac{C_Q}{\varepsilon_{\text{SC2}}} P^{1/2} (\log P)^C \right\rceil.$$

Equivalently, one may use the asymptotic form

$$H(P) \geq H_{\min}^{\text{asym}}(P) := \left\lceil P^{1/2}(\log P)^{C_*} \right\rceil, \quad C_* > C.$$

Under either threshold, the same conditional contradiction-field chain gives an element

$$x \in I, \quad p \nmid x(x-w) \quad (p \leq P).$$

When $x - w > 1$, both x and $x - w$ are primes.

Recorded scale check. The only scale-sensitive term in the recorded chain is the middle/high-scale $K = 2$ near-crossing error. With top layer $Y = P^{3/4}$ and $Q_{\text{eff}} \leq (\log P)^C$, replacing the original back-half length $\asymp P^2$ by $|I| \asymp H(P)P$ changes the normalized error to

$$\frac{Q_{\text{eff}} Y^2}{|I|} \ll \frac{P^{1/2}(\log P)^C}{H(P)}.$$

The displayed lower bound on $H(P)$ makes this at most ε_{SC2} in the explicit form, or $o(1)$ in the asymptotic form. Hence the constants in the TCA, SC2, zero-mass, finite-package, and smoothing ledgers are unchanged. The conclusion is therefore a conditional window-compressed version of the previous proposition. It becomes an unconditional prime-pair theorem only if I3-Core is independently proved and accepted. $\square$

Remark 2.10 (Referee boundary). The two-point secondary-sieve chapter is included to document the contradiction-field architecture and its parameter ledger. The window-compression audit is recorded in `docs/monograph/two-point-window-compression-and-unconditionality.md`. It must not be cited as an unconditional proof of prime pairs, the twin-prime conjecture, or Goldbach-type assertions unless I3-Core is independently proved and accepted.

Remark 2.11 (Current obstruction). The current hard attack reduces I3-Core to a true-residual correlation problem. The local CRT and matrix rigidities do not by themselves exclude a model residual set concentrated on a new prime's bad residue class while still satisfying all previously imposed two-residue exclusions and short-window non-reuse constraints. Thus the remaining task is not another bookkeeping step but a genuine lower-bound/correlation theorem for the true residual set in new prime moduli.

Remark 2.12 (Latest TRC-1G reduction). The single-prime part of the remaining TRC input has been sharpened to a heavy-atom prohibition. If more than 40% of the true residual mass lies in the two bad classes $0, w \pmod{p}$ for some new prime $p > Y$, then Parseval gives a non-zero Fourier coefficient modulo p of size $> 1/5$ after normalization. Equivalently, the three long-direction difference families $0, \pm w \pmod{p}$ carry a fixed positive proportion of the true-residual pair energy. The existing matrix rigidities remove the short-direction, 45°-locked, reuse/shared-complement, and endpoint pieces. The remaining unproved input is therefore a new-modulus long-direction balance theorem for the true residual set, not a finite-template or local CRT bookkeeping issue.

Remark 2.13 (Total large-incidence reduction). The pointwise new-modulus balance can be weakened further. It is enough to prove the total large-incidence inequality

$$\sum_{P^\alpha < p \leq P} \#\{x \in U_{P^\alpha}(I) : p \mid x(x-w)\} < |U_{P^\alpha}(I)|.$$

Indeed, a point not counted on the left is unsieved by every prime up to P , and in the P^2 matrix range this forces both x and $x - w$ to be prime. The CRT model predicts the normalized left side to be $2 \log(1/\alpha) + o(1)$; for $\alpha = 3/4$ this is $2 \log(4/3) = 0.57536 \dots$, leaving margin $0.42463 \dots$. The unproved part is the true-residual total-incidence estimate with an error below that margin.

Remark 2.14 (RB–TLI audit boundary). The direct attempt to prove the total large-incidence estimate separates cleanly into a numerator and a denominator. The numerator is an upper-sieve problem: after writing $x = pm$ or $x = w + pm$, the variable m again avoids two residue classes modulo each old prime, giving an upper bound of shape

$$A_p \leq 2C_+ |I| V_2(P^\alpha) / p + R_p.$$

The denominator requires a model-level lower bound $|U_{P^\alpha}(I)| \geq c_- |I| V_2(P^\alpha)$. For $\alpha = 3/4$, this route would need $C_+/c_- < 1/(2 \log(4/3)) = 1.738 \dots$. This relative Buchstab total-incidence input is the remaining unproved prime-pair-strength statement.

Remark 2.15 (Local collapse when $q \mid w$). If an old odd prime q divides the fixed even difference w , then the two forbidden classes $0, w \pmod{q}$ coalesce into a single forbidden class. The local density is therefore $1 - 1/q$ instead of $1 - 2/q$, giving the singular local gain $(q-1)/(q-2)$. This increases the absolute number of admissible residual points and weakens the small-prime synchronization peaks. In the relative RB–TLI ratio, however, the same local factor multiplies both $|U_Y|$ and the model numerator $A_p \sim 2|I|V_w(Y)/p$ for all new primes $p > Y > |w|$. Thus the main relative constant remains $2 \log(1/\alpha)$; the collapse improves finite constants but does not by itself close the hardest case $w = 2$.

Remark 2.16 (Buchstab semiprime transition for $w = 2$). The latest numerical audit of the hardest case $w = 2$ shows that the naive constant $2 \log(1/\alpha)$ is not the correct conditional main term after the old primes have been sieved. For $\alpha > 2/3$, any new-prime hit $p \mid x$ or $p \mid x - 2$ with $Y = P^\alpha < p \leq P$ forces the complementary quotient to be prime, because it is Y -rough and smaller than Y^2 . Thus the large-incidence layer is a Buchstab semiprime transition. The corresponding two-sided main constant is

$$K(\alpha) = \frac{2 \log((2 - \alpha)/\alpha)}{1 + \log((2 - \alpha)/\alpha)}.$$

Exact scans recorded in `docs/rb-tli-w2-scan-large.md` give, for example, $E_U D = 0.691696$ at $P = 10007, \alpha = 3/4$, compared with $K(3/4) = 0.676220$. The remaining target is therefore a two-point stability theorem for this Buchstab semiprime transition, not the earlier naive incidence model.

Remark 2.17 (BST error decomposition). The refined scan decomposes the remaining error. At $P = 10007, \alpha = 3/4$, the shell contributions in the five exponent intervals between 0.75 and 1 track the Buchstab shell constants, and the covariance between the x -side and $(x-2)$ -side large-incidence counts is approximately $-9.3 \cdot 10^{-5}$. The zero-incidence set is independent of the intermediate cutoff α , as it is exactly the set unsieved by all primes up to P . Thus the remaining hard input has been reduced to a shifted biprime roughness estimate of the form

$$\#\{p, m : pm \in I, pm - 2 \in U_Y\} \leq (B(\alpha) + o(1))|U_Y|.$$

This is the current BST–2 obstruction.

Remark 2.18 (BMD reduction). The BST–2 obstruction can be written as a one-dimensional multiplicative sieve on biprime variables. After setting $x = pm$, the condition $pm - 2$ is unsieved by old primes is, for each $q \leq Y$,

$$pm \not\equiv 2 \pmod{q}.$$

The prime $q = 2$ is peeled off deterministically: for odd primes p, m , the integer $pm - 2$ is odd. Thus the multiplicative sieve is taken over odd q . For each odd $q \leq Y$, since $p, m > Y \geq q$, the forbidden

curve has $(q-1)$ points inside $(\mathbb{F}_q^\times)^2$, out of $(q-1)^2$ possible non-zero pairs. The local forbidden density is therefore $1/(q-1)$. The remaining unproved estimate is a biprime multiplicative dispersion bound: weighted sums of the errors in the congruences $pm \equiv 2 \pmod{d}$, with odd squarefree d , must be $o(|U_Y|)$ for the Buchstab/Rosser weights used in the transition. This is the current BMD input.

Remark 2.19 (Row-column input for BMD). If the prime-matrix row/column theorem is accepted as a conditional input, it can be used only to remove zero-section degeneracies in complete CRT blocks: entire empty rows, columns, or unit residue cells are then not allowed to masquerade as the main BMD error. It does not by itself prove BMD, because BMD is a signed distribution statement. For odd d , expanding

$$1_{pm \equiv 2 \pmod{d}} = \frac{1}{\varphi(d)} + \frac{1}{\varphi(d)} \sum_{\chi \neq \chi_0} \bar{\chi}(2) \chi(p) \chi(m)$$

shows that the remaining obstruction is a bilinear character-dispersion estimate

$$\sum_{d \leq D} \frac{|\lambda_d|}{\varphi(d)} \sum_{\chi \neq \chi_0} \left| \sum_{Y < p \leq P} \chi(p) \sum_{\substack{m \in J_p \\ m \text{ prime}}} \chi(m) \right| = o(|U_Y|).$$

Thus the row/column input gives a useful BMD-Zero exclusion, but the genuine remaining hard point is BMD-Char.

Remark 2.20 (Direct BMD attack via an E_2 Bombieri–Vinogradov input). Set $N \asymp P^2$ and

$$a_n = \#\{(p, m) : Y < p \leq P, m \text{ prime}, n = pm, n \in I\}.$$

For $\alpha > 2/3$ the multiplicity of a_n is bounded. After the deterministic modulus 2 has been peeled off, the BMD remainder for odd squarefree d is exactly the arithmetic-progression discrepancy of this restricted two-prime convolution:

$$R_d = \sum_{\substack{n \in I \\ n \equiv 2 \pmod{d}}} a_n - \frac{1}{\varphi(d)} \sum_{\substack{n \in I \\ (n, d) = 1}} a_n.$$

Thus BMD follows from the standard Bombieri–Vinogradov theorem for restricted E_2 sequences at level

$$d \leq P(\log P)^{-B} = N^{1/2}(\log N)^{-B+O(1)}.$$

Indeed, if

$$\sum_{d \leq P/\log^B P} \max_{(a, d) = 1} |R_d(a)| \ll_A N(\log N)^{-A},$$

then every Rosser/Buchstab weighted remainder is $o(|U_Y|)$ since $|U_Y| \asymp N/(\log P)^2$. The condition $\alpha < 1$ ensures $Y < P/\log^B P$ for large P , so all one-prime forbidden curves $q \leq Y$ are included in the available distribution range. Therefore BMD is closed if this standard E_2 BV input is cited; a fully self-contained manuscript must instead prove that input in an appendix.

Remark 2.21 (Self-contained BV- E_2 obligation). The local BV- E_2 appendix records the exact self-contained obligation. A direct multiplicative large-sieve estimate is insufficient: in the balanced dyadic block $p \sim P_1$, $m \sim M_1$, with $P_1 \asymp M_1 \asymp P$ and $Q \asymp P/\log^B P$, Cauchy plus the large sieve gives size roughly $P^3/\log^{2B} P$, while the required Bombieri–Vinogradov error is $P^2/\log^A P$. Thus the only genuinely deep remaining self-contained input is a Type-II dispersion/Kloosterman-cancellation estimate for the balanced convolution blocks. If that standard dispersion theorem is cited, the BMD step is closed as an external-input theorem; if not, this is the next appendix that must be proved line by line.

Remark 2.22 (Weighted BE2-3K core). BMD does not require the full $\max_a$ Bombieri–Vinogradov statement. It only needs the fixed class $2 \bmod d$ against the well-factorable Rosser/Buchstab weights. After Cauchy, the self-contained problem is a variance bound for

$$T_r = \sum_{d \leq Q} \lambda_d \left(\sum_{s \equiv 2r^{-1} \bmod d} \beta_s - \frac{1}{\varphi(d)} \sum_{(s,d)=1} \beta_s \right).$$

Opening $\sum_r |T_r|^2$, the off-diagonal conditions

$$rs_1 \equiv 2 \pmod{d_1}, \quad rs_2 \equiv 2 \pmod{d_2}$$

give, by CRT and Fourier expansion of the interval condition, the Kloosterman phase

$$e\left(-\frac{2h\overline{s_1}\overline{d_2}}{d_1} - \frac{2h\overline{s_2}\overline{d_1}}{d_2}\right).$$

Thus the true no-black-box core is a weighted bilinear Kloosterman dispersion estimate, denoted BE2-3K in the notes. Proving BE2-3K gives BE2-3, then weighted BV- E_2 , then BMD.

Remark 2.23 (Referee status of the no-black-box claim). At the current stage the manuscript is not yet fully no-black-box. The BMD obstruction has been reduced to the single deep BE2-3K Kloosterman-dispersion estimate. If a BFI/Deshouillers–Iwaniec/Kuznetsov type mean-value theorem is cited, BMD is closed as an external-input theorem. If no external input is allowed, BE2-3K remains the unique missing line-by-line proof obligation.

Remark 2.24 (Further reduction of BE2-3K). The BE2-3K core can be narrowed further. Point-wise Weil bounds for individual incomplete reciprocal sums give only single-modulus square-root cancellation and do not provide the arbitrary logarithmic saving required after summing over the d_1, d_2, h families. The common-divisor strata $(d_1, d_2) > 1$ are not the main obstruction: compatibility forces $s_1 \equiv s_2$ modulo the common divisor and yields only polylogarithmic loss. The remaining genuine input is a windowed Kloosterman spectral large-sieve estimate, denoted KLS-window in the notes. Its role is precisely

$$\text{KLS-window} \Rightarrow \text{BE2-3K} \Rightarrow \text{BE2-3} \Rightarrow \text{weighted BV-}E_2 \Rightarrow \text{BMD}.$$

Remark 2.25 (External-theorem closure of KLS-window). The KLS-window input is tied to two classical external sources: the Deshouillers–Iwaniec spectral Kloosterman large-sieve technology, as developed in *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. 70(2), 219–288 (1982), and the Bombieri–Friedlander–Iwaniec dispersion framework with well-factorable weights, as in *Primes in Arithmetic Progressions to Large Moduli. II*, Math. Ann. 277, 361–394 (1987). With these external theorems allowed, the analytic chain closes as

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2-3K} \Rightarrow \text{BE2-3} \Rightarrow \text{weighted BV-}E_2 \Rightarrow \text{BMD}.$$

This is an external-deep-theorem closure, not a fully self-contained reproof of Kuznetsov/dispersion theory.

Chapter 3

Zeta-Zero Contradiction Fields

Assume an off-critical zero $\rho = \beta + i\gamma$ with $\beta > 1/2$. The RH contradiction-field manuscript transfers the resulting smooth prime anomaly into a CRT rough-composite ledger, decomposes it into sparse/dense/tail/internal/global exits, and proves that no named escape channel remains under the recorded local theorems and restricted external inputs.

Theorem 3.1 (RH contradiction-field synthesis, recorded form). *Under the numbered local theorems, controlled-exit lemmas, threshold hierarchy, and restricted external inputs recorded in `paper/rh-proof/rh-contradiction-field.tex`, an off-critical zero has no free escape channel in the contradiction-field event graph.*

Reference. See the RH manuscript in `paper/rh-proof/` and the final audit in `docs/`. The submission warning in the RH manuscript remains in force until independent line-by-line referee verification and final monograph page-number checks are completed. $\square$

Chapter 4

Unified Dependency Map

The two programs share the same logic:

counterexample $\Rightarrow$ structured covering field $\Rightarrow$ named exits $\Rightarrow$ capacity/no-cycle contradiction.

The prime-matrix program is more geometric and finite-CRT in nature; the zeta-zero program is analytic and scale-asymptotic. The monograph treats both as projections of the same contradiction-field method.

Layer	Prime-matrix projection	RH projection
Baseline	CRT nonzero class balance	smoothed CRT candidate baseline
Local rigidity	large-factor non-reuse, diagonal locks	sparse/dense/tail routing
Capacity	anchor coverage capacity	projection and square-function capacity
No-cycle	recursive peeling and shell migration	GEE transfer/no-cycle ledger
External inputs	finite verification and Structured-EHPD	explicit formula, KL, Vaaler, BG/Baker, sieve, Vaughan
Final status	reduction package, not final unconditional theorem	verification package with submission warning

Chapter 5

Directory and Theory System

The detailed working directory and theory-system map is maintained in

`docs/monograph/combined-monograph-directory-and-theory-system.md`.

This chapter records the concise version used by the merged monograph.

5.1 Proposed Book Structure

Part	Contents	Status
I	Unified contradiction-field method: entrance, local rigidity, exits, capacity, no-cycle potential	Methodological framework
II	Prime-matrix row/column program: CRT skeleton, A/B reduction, Structured-EHPD interface	Reduction package; not final theorem
III	Two-point secondary sieve: TLI, BST, BMD, WBE2, BE2-3K, KLS-window	External-deep-theorem closure for BMD
IV	Zeta-zero contradiction field: explicit-formula entrance and controlled exits	Verification package; not final RH theorem
V	Dependency map, referee checklist, optimization routes	Audit and future work

5.2 Latest Two-Point Chain

For the $w = 2$ hardest two-point branch, the current analytic chain is

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2-3K} \Rightarrow \text{BE2-3} \Rightarrow \text{WBE2} \Rightarrow \text{BMD} \Rightarrow \text{BST-2} \Rightarrow \text{BST} \Rightarrow \text{TLI}.$$

The chain is closed only in the external-deep-theorem sense: DI and BFI are cited as classical analytic inputs. A fully self-contained version would require reproving the spectral Kloosterman large-sieve and dispersion machinery.

5.3 Prime-Matrix Low-Hole Closure Package

The latest prime-matrix update closes the low-hole bucket submodule as a finite-certificate plus explicit-external-theorem package. The verification is split into five non-overlapping ranges:

$$\begin{aligned}
 13 \leq P < 61 & : \text{low-range final certificate,} \\
 61 \leq P \leq 103 & : \text{narrow-band collision-energy certificate,} \\
 107 \leq P \leq 229 & : \text{exact } P/5 \text{ splitting recursion,} \\
 233 \leq P \leq 13207 & : \text{integer cross-multiplied product certificate,} \\
 P \geq 13208 & : \text{Rosser–Schoenfeld explicit constants.}
 \end{aligned}$$

The low-range certificate checks 15414 zero buckets: 15282 close by whole-set Hall deficit, 108 close by bridge-critical deletion, and the remaining 24 exceptional phases at $P = 41$ close by exact finite set-cover dynamic programming. The narrow band $61 \leq P \leq 103$ covers 23100 low phases by fixed ascending high-prime residue blocks. The two finite tail certificates cover 23 and 1520 prime rows respectively, the latter by exact integer cross multiplication rather than floating-point comparison.

For the infinite tail, Rosser–Schoenfeld Corollary 1, formulas (3.5)–(3.6), supplies the prime-counting bounds, while Theorem 7, formula (3.26), supplies the Mertens product bound. Since $P \geq 13208$ implies $\lfloor P/5 \rfloor \geq 2641$, the factor $1 + 1/(2 \log^2 \lfloor P/5 \rfloor)$ is below 1.009, hence the manuscript’s conservative 1.03 Mertens constant is admissible.

This closes the BPN low-hole bucket interface in the stated finite-certificate/external-theorem sense. It does not by itself promote the whole prime-matrix row/column program to a final theorem, because the remaining global exits still require the separately named PDEC/SAE and Rankin-ledger acceptance obligations.

5.4 Claim Status Legend

The monograph uses the following status labels.

Label	Meaning
Proved-in-text	The statement is proved inside the manuscript under the stated hypotheses.
Reduction-closed	The statement has been reduced to a smaller named interface.
External-theorem closed	The statement is closed after citing explicitly named external theorems.
Computational-certificate	The statement is closed over a finite range or certified submodule by reproducible scripts and archived outputs.
Referee-block	A proof package exists, but it still requires line-by-line referee verification.
Not claimed	The statement is not asserted as a theorem of the manuscript.

The status discipline and promotion rules are maintained in

`docs/monograph/claim-status-discipline.md.`

The manuscript provides theorem-like environments with matching names for future editing. A statement may be promoted only when its required proof, reduction endpoint, external theorem template, or certificate package is present.

5.5 External Theorem Index

The external-input index is maintained in

`docs/monograph/external-theorem-index.md`.

For the prime-matrix low-hole tail, the current explicit package is Rosser–Schoenfeld 1962: Corollary 1, formulas (3.5)–(3.6), supplies the required $\pi(x)$ bounds, and Theorem 7, formula (3.26), supplies the required Mertens product bound. These constants close the $P \geq 13208$ tail in the external-theorem sense once the finite certificates below that threshold are accepted.

For the two-point sieve, the critical external package is DI plus BFI. The index records the required checks for the KLS-window input: Kloosterman phase normalization, modulus range, frequency range, inverse-variable support, well-factorable weights, gcd strata, smoothing and sawtooth losses, and the final choice of the logarithmic-loss parameter $B(A)$. The standalone adaptation template is maintained in

`docs/monograph/kls-window-di-bfi-adaptation-template.md`.

It closes the BMD distribution input only in the external-deep-theorem sense; the separate BMD-to-TLI transfer remains a hard obligation.

5.6 Critical Chain Audit

The detailed chain audit is maintained in

`docs/monograph/key-proof-chain-optimization-audit.md`.

Its main purpose is to prevent status mixing. Prime-matrix existence rigidity may support zero-section exclusion, but it does not replace signed spectral distribution estimates such as BMD-Char. Numerical experiments may guide constants and structures, but they are not proof inputs. The RH branch remains a verification package and is not claimed as an unconditional proof.

5.7 Optimization Priorities

1. Separate theorem environments for proved-in-text, reduction-closed, external-theorem closed, and referee-block statements.
2. Replace the overly strong $\max_a \text{BV-}E_2$ formulation by the precise WBE2 well-factorable weighted statement in the main line.
3. Rewrite KLS-window directly in the notation of the cited DI/BFI theorems to reduce translation risk.
4. Compress the prime-matrix program around the single Structured-EHPD exclusion interface.
5. Preserve the RH warning language until every controlled exit has independent referee verification.

5.8 Pre-External-Referee Hard Obligations

The current hard-obligation ledger is maintained in

`docs/monograph/pre-external-referee-remaining-obligations.md`.

It records the author-side status of every terminal interface that cannot honestly be promoted to a final theorem without further proof or independent verification. The decisive interfaces are:

Program	Hard interface	Required closure before theorem promotion
Prime matrix	Structured-EHPD / D-structure exclusion	Prove the exclusion theorem directly, or replace it by a fully proved RHI-type large-prime incidence inequality.
Prime matrix	PDEC / SAE / Rankin exits	Prove persistent endpoint CRT defect exclusion, local isolated-escape exclusion, and complete the formal Rankin certificate ledger.
Prime matrix	RRD / OSPC / Selberg-uniform constants	Put all constants in one convention and prove each budget item over the full asymptotic range, not only in sampled windows.
Two-point sieve	I3-Core true-residual package	Prove true-residual new-modulus balance, TCA, SC2, 45° synchronization peak control, Directional Balance, Zero Mass, and endpoint smoothing.
Two-point sieve	DI/BFI to KLS-window adaptation	State the external theorems in their standard variables and verify phase, modulus, frequency, weights, gcd strata, and logarithmic-loss absorption.
Two-point sieve	BMD to TLI transfer	Prove the Buchstab transfer and total-large-incidence comparison without using a hidden prime-pair lower bound.
RH branch	controlled exits	Prove every sparse, dense, tail, internal, analytic, and global exit under a single load convention.

For the RRD/OSPC constants, the current acceptance matrix is maintained in

`docs/monograph/h5-rrd-ospc-proof-obligation-matrix.md`.

It splits the remaining constant obligation into six independently checkable inequalities and does not claim that those inequalities have already been proved. The first item, RRD-low routing, is recorded in

`docs/monograph/h5-1-rrd-low-exit-theorem.md`.

It proves that an RRD-low excess enters either OSPC* or a weighted CRT-defect exit; excluding those exits remains part of the PDEC/SAE and tail-anchor obligations. The absorption of these two exits into the unified PDEC-or-SAE interface is recorded in

`docs/monograph/h5-4-ospc-weighted-crtdefect-absorption.md.`

This absorption is not an exclusion theorem; the required PDEC and SAE certificates remain open hard obligations. The PDEC certificate format itself is now recorded in

`docs/monograph/h4-pdec-certificate-template.md.`

It fixes the data, Fourier lower bound, admissible CRT upper-bound certificates, and failure-routing rules for persistent defects. This is a certificate-format closure only; the actual PDEC certificates for the formal bad-window families remain to be supplied. The first source lemmas for admissible PDEC constraint rows are recorded in

`docs/monograph/h4-pdec-constraint-source-lemmas.md.`

They justify mass, phase-capacity, mirror-pair, low-hole-bucket, and conditional-routing rows when their stated hypotheses hold. They do not by themselves provide the full constraint table or the final dual upper bound. The first admissibility table for those rows is recorded in

`docs/monograph/h4-pdec-admissible-constraint-table.md.`

It separates tautological, finite-certificate, symbolic, conditional-routing, still-unproved, and rejected rows. In particular, full-cycle CRT balance is not allowed to control an arbitrary bad-window subset. The column-cap source rule is recorded in

`docs/monograph/h4-pdec-column-cap-source-lemma.md.`

It allows column information to enter only through finite column-projection capacity, symbolic column-capacity theorems, or conditional routing to ColumnRadius/ColumnCRT/Tail-anchor exits. The coefficient ledger for those rows remains open. The first coefficient ledger is recorded in

`docs/monograph/h4-pdec-column-cap-coefficient-ledger.md.`

It registers finite column-witness, RCI/CDB, LHB column-residue, and conditional ColumnDefect rows. Most rows still require machine-readable phase blocks before they can become actual rows of the dual certificate matrix. The first materialized phase-block file is

`docs/monograph/h4-pdec-lhb-column-phase-blocks.json.`

It gives finite $Q = 2310$ LHB column rows for $13 \leq p \leq 47$. Empty anomaly blocks are machine-readable finite rows; diagnostic support blocks still require a projection-capacity proof before entering the final dual system. The support-to-capacity transfer conditions are recorded in

`docs/monograph/h4-pdec-lhb-support-to-capacity-transfer.md.`

They make explicit that support size is not a multiplicity bound for the persistent count vector $g(t)$; a phase-indicator reduction, a multiplicity cap, or an allowed-set projection bound is still required. The multiplicity route is made explicit in

`docs/monograph/h4-pdec-lhb-multiplicity-cap-route.md.`

It states the precise acceptance contract: for the same (p, Q, S, τ) one must prove $g(t) \leq M(t)$, and only then may a diagnostic block C be promoted to the capacity row $\sum_{t \in C} g(t) \leq \sum_{t \in C} M(t)$. The first finite multiplicity-cap certificate is

`docs/monograph/h4-pdec-lhb-multiplicity-cap-certificate.json.`

It takes $M(t)$ to be the high-layer CRT completion count $C_P(t; Q)$ and verifies, for $Q = 2310$ and $p = 13, 17, 19, 23, 29, 31, 37, 43, 47$, that the WHOLEDEF and BRIDGED support blocks have $\sum_{t \in C} M(t) = 0$. This is an allowed-set projection certificate: it can be used in the LHB branch after proving $S \subset Z_{\text{LHB}}(p, Q)$, but it is not by itself a global PDEC exclusion theorem. The attachment lemma for that branch is

`docs/monograph/h4-pdec-lhb-attachment-lemma.md.`

It proves that an LHB-typed full-row bad window is indeed an element of $Z_{\text{LHB}}(p, Q)$. Thus the remaining global task is no longer this attachment step, but the classification step: every formal PDEC bad window must either be LHB-typed or be routed to a named SAE, ColumnCRT/ColumnRadius, TailAnchor, or Rankin exit. The classification step is recorded in

`docs/monograph/h4-pdec-bad-window-classification-lemma.md.`

It combines the unified PDEC-or-SAE dichotomy, column-conflict routing, tail-anchor persistence, and Rankin access rules to split each named low-mod bad-window family into SAE, LHB-PDEC, or Routed-PDEC. This is still a routing theorem: the non-LHB exits and the final dual $U_{\text{CRT}} < L_{\text{PDEC}}$ certificate remain open. The homogeneous-splitting rule is recorded in

`docs/monograph/h4-pdec-homogeneous-splitting-lemma.md.`

It closes the bookkeeping case in which one attempted certificate mixes different (Q, τ, F, κ, p) types: such a family must be partitioned into homogeneous subfamilies before applying UPS-1 or the bad-window classification. Thus mixed type is not an additional mathematical exit; the remaining exits are SAE, ColumnCRT/ColumnRadius, TailAnchor, and Rankin. The column-defect routing contract is recorded in

`docs/monograph/h4-pdec-column-defect-routing-contract.md.`

It defines the conditional certificate objects for ColumnRadius and ColumnCRT rows: radius rows use a threshold D_0 , a column-witness selector, and phase-compatible weights $W_D(t)$, while displacement rows use a label ℓ , a non-zero displacement residue a , a threshold L_D , and phase-compatible weights $W_{\ell,a}(t)$. Violating such a row routes to the named ColumnRadius or ColumnCRT exit after the zero displacement class has been separated by the same-column prime-witness rigidity. This is a routing-contract closure only; the exits themselves still require materialized weights, thresholds, and exclusion certificates. The first finite materialization of those weights is recorded in

`docs/monograph/h4-pdec-column-defect-weight-certificate.json.`

It uses the refined finite phase $\tau_{\text{fin}} = (p, q, \text{row})$ for $p \leq 1000$ and the five tightest RCI rows for each p . In this finite domain, the blocks $D_{\text{col}} > 81$ and displacement-residue load > 2 are both empty, over 835 phases. This is a finite refined-phase certificate; it does not provide a global D_0, L_D theorem or exclude the ColumnRadius/ColumnCRT exits. The recursive zero-row peeling route is isolated in

`docs/monograph/prime-matrix-recursive-peeling-zero-row-hardpoint.md.`

It proves the exact layer-peeling identity: after a p -sieve zero window is peeled to the previous prime level, any resurrected survivor must be a multiple of the peeled prime with rough cofactor. Thus the recursive object is a punctured zero window, not automatically a lower-level zero row. The accompanying finite audit finds no forced consecutive lower zero rows in the known first-zero-row samples. The useful remaining target is therefore an absorption theorem routing those resurrected points to ColumnCRT, ColumnRadius, or TailAnchor, rather than a direct short-period contradiction. The scaled-row and half-width version is audited in

`docs/monograph/prime-matrix-scaled-peeling-halfwidth-audit.md.`

It shows that the scaled height nP/p is only an approximate locator: a lower aligned zero row appears only when the upper window fully contains a lower row. In the same samples, passing to a prime near half the upper width gives no automatic pair of consecutive lower zero rows, because the half-level sieve resurrects points whose least factor lies above the half-width level. The resurrected-point absorption route is now recorded in

`docs/monograph/prime-matrix-rpz-absorption-defect-route.md`

with the finite audit

`docs/monograph/prime-matrix-rpz-absorption-defect-audit.md.`

It routes complete absorption into five alternatives: a survivor remains, TailAnchor load, ColumnCRT displacement load, ColumnRadius, or a residual Distributed-RPZ branch. In the known first-zero-row samples the half-level has 15 resurrected points, no missing absorber, and maximum one-window label and top-column loads equal to 1. Thus the route does not justify a single-window TailAnchor exclusion; the next theorem-level gate is the Distributed-RPZ capacity barrier, or a proof that distributed absorption necessarily returns to one of the named column or tail exits. The first attack on this gate is recorded in

`docs/monograph/prime-matrix-rpz-sliding-plateau-barrier.md.`

It proves a sliding-platform multiplicity formula: if a zero window remains zero under a consecutive block of shifts, then a resurrected source in the common core repeats with the full platform multiplicity. Hence distributed absorption on such a platform is either a persistent TailAnchor load or, after source deletion, a boundary-compressed branch. The audit in

`docs/monograph/prime-matrix-rpz-sliding-plateau-audit.md`

finds platform length 5 in the known samples and TailAnchor triggering at $T_0 = 4$ in all five cases. The remaining gate is therefore the boundary-compressed RPZ branch, not unrestricted distributed absorption. The boundary-compressed branch is further routed in

`docs/monograph/prime-matrix-rpz-boundary-compressed-core-route.md.`

It proves that, if TailAnchor is excluded, the center interval $J_{T_0} = [L + A + T_0, R + B - T_0]$ is an h -sieve zero interval. The audit

`docs/monograph/prime-matrix-rpz-bcb-core-audit.md`

finds that all 11 center survivors in the known samples are TailAnchor core sources; after conditional TailAnchor deletion, all five center intervals are clean and contain an aligned half-width row. The

next gate is therefore the BCB grid/endpoint exclusion: either the center contains an aligned lower zero row, or the endpoint seam enters SAE, PDEC, or ColumnCRT.

These obligations are not typographical tasks. They are the remaining theorem-level gates. Until they are closed, the monograph may be submitted only as a synthesis, dependency, and verification manuscript, not as a final unconditional proof of the prime-matrix theorem, prime-pair assertions, or RH.

Chapter 6

Referee Closure Audit

This chapter records the strict closure standard used for the whole monograph. A contradiction-field chain is considered closed only if its entrance, decomposition, exits, and global contradiction are all verified under the same load convention.

Definition 6.1 (Closure certificate). A closure certificate for a contradiction field consists of four data:

1. an entrance theorem sending every counterexample into the field;
2. a disjoint or source-deleting decomposition into named exits;
3. for each exit, an absorption, descent, transfer, external theorem, or finite certificate;
4. a global comparison showing that final load is simultaneously bounded below and above in incompatible ways.

If any item is replaced by an unverified structural input, the conclusion is conditional on that input.

Theorem 6.2 (Current monograph closure). *The present manuscript closes the audit and dependency structure of the two contradiction-field programs: all known entrances, decompositions, exits, external inputs, finite certificates, and referee obligations are explicitly named. It does not close the final unconditional theorem status of either program.*

Proof. For the prime-matrix program, the A/B appendix gives the entrance reduction from a row or column counterexample to a Structured-EHPD bad configuration. Inside the BPN branch, the low-hole bucket submodule is now closed by the low-range certificate, the narrow-band collision certificate, the two finite tail certificates, and the Rosser–Schoenfeld explicit tail package recorded above. This is a genuine submodule closure. The D-structure exclusion and the remaining PDEC/SAE/Rankin global exits remain decisive inputs requiring independent line-by-line review. Therefore the whole chain is closed as a reduction and certificate package but not as a final theorem.

For the RH program, the RH manuscript has completed U1–U5: statement wording, AEX acceptance, EXT localization, manuscript wording, BibTeX/PDF compilation, and warning-free LaTeX logs. The final audit still requires independent referee verification of every local theorem and controlled exit. Therefore the chain is closed as a verification package but not as a final unconditional RH theorem.

The status table and the referee audit in `docs/monograph/` record exactly which pieces are proved, reduced, external, computational, or awaiting review. Thus the monograph’s logical boundary is explicit and closed, while promotion to final theorem status remains conditional on external verification. □

Theorem 6.3 (Internal line-by-line audit). *The author-side line-by-line audit has found no new item classified as a mathematical block. The remaining blocks are referee blocks: Prime Matrix final promotion requires independent acceptance of the D-structure/Tail-log4/finite-verification interface, and RH final promotion requires independent acceptance of all controlled exits in the RH contradiction-field manuscript.*

Proof. The internal audit matrix assigns one of four labels to each link. No link is a math-block.

The terminal entries PM-16 and RH-16 remain referee-blocks. An author-side check cannot replace independent referee verification. Therefore the internal audit is complete, but final unconditional theorem promotion is not. $\square$

Remark 6.4 (No overclaim principle). The monograph may state the shared contradiction-field method and the completed audit closures. It must not state unconditional proofs of RH or of the prime-matrix row/column theorem until the remaining structured inputs have independent acceptance.

Chapter 7

Current Status and Honest Boundary

The purpose of this monograph draft is to make the shared structure visible and auditable. It is not yet a final unconditional theorem manuscript. The precise statuses are maintained in `docs/monograph/claim-status-table.md`. Any future promotion to final theorem status requires independent verification of every reduced structured input and every controlled exit.